METRIC
MIL-PRF-55365/8J
w/AMENDMENT 1
13 March 2020
SUPERSEDING
MIL-PRF-55365/8J
5 September 2014

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, FIXED, TANTALUM, POLARIZED, ESTABLISHED RELIABILITY, NON-ESTABLISHED RELIABILITY AND HIGH RELIABILITY, STYLE CWR11 (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55365.

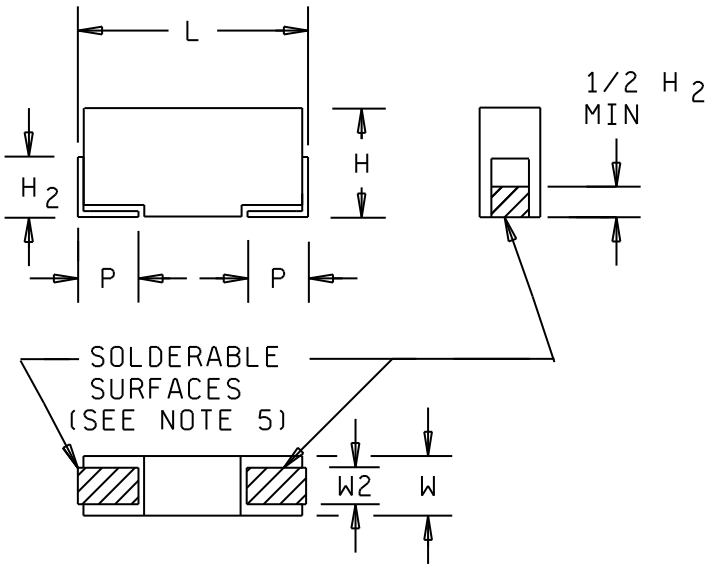


FIGURE 1. Style CWR11 (molded) capacitors.

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Dimensions						
Case code	H	H_2 (min)	L	P ±0.3 (.012)	W	W_2 ±0.1 (.004) (see note 4)
A	1.6 ±.2 (.063 ±.008)	0.7 (.028)	3.2 ±.2 (.126 ±.008)	0.8 (.031)	1.6 ±.2 (0.063 ±.008)	1.2 (.047)
B	1.9 ±.2 (.075 ±.008)	0.7 (.028)	3.5 ±.2 (.138 ±.008)	0.8 (.031)	2.8 ±.2 (.110 ±.008)	2.2 (.087)
C	2.5 ±.3 (.098 ±.012)	1.0 (.039)	6.0 ±.3 (.236 ±.012)	1.3 (.051)	3.2 ±.3 (.126 ±.012)	2.2 (.087)
D	2.8 ±.3 (.110 ±.012)	1.0 (.039)	7.3 ±.3 (.287 ±.012)	1.3 (.051)	4.3 ±.3 (.169 ±.012)	2.4 (.094)

NOTES:
1. Dimensions are in millimeters. Inch equivalents are given in parenthesis for general information only.
2. These capacitors are designed for mounting by dip or wave soldering, reflow soldering, thermo-compression bonding, or other conventional means.
3. The anode (+) terminal shall be indicated by a marking on the case or by bevel on anode end.
4. Applicable to solderable surfaces only. The configuration of the nonsolderable area is optional.
5. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
6. For solder coated terminations, add an additional 0.38 mm (.015 inch) to the tolerances for “L”, “H”, “P”, and “W_2” for each case size.

FIGURE 1. Style CWR11 (molded) capacitors - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Termination finish: In accordance with MIL-PRF-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365.

DC leakage (DCL): See table I.

Capacitance: See table I.

Capacitance tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365. See table I.

Resistance to soldering heat: In accordance with MIL-PRF-55365.

Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Life:

2,000 hours: In accordance with MIL-PRF-55365.

10,000 hours: In accordance with MIL-PRF-55365.

Solderability: In accordance with MIL-PRF-55365, except that following steam aging; test samples may have a 30 minute bake out at +150°C prior to solder dipping.

Marking: In accordance with MIL-PRF-55365, to be marked on top of case as follows:

Case code A

J 105X	(+) -	polarity stripe, J = JAN capacitance in picofarad code and manufacturer's identifier

Case codes B, C, and D

J 105 35X	(+) - -	polarity stripe, J = JAN capacitance in picofarad code rated voltage and manufacturer's identifier

Alternate case code A marking

X 105J	(+) -	polarity stripe, X = manufacturer's identifier capacitance in picofarad code; J = JAN

Print orientation is optional. Additional marking may appear provided that it does not interfere with the required marking. At the option of the manufacturer, the lot date code may be marked on the top of the case for any case code.

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TABLE I. Style CWR11 characteristics.

Part or identifying number (PIN) 1/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR11C-225 - - -	4	2.2	0.5	5.0	6.0	6	9	9	8.0	A
CWR11C-475 - - -	4	4.7	0.5	5.0	6.0	6	9	9	8.0	A
CWR11C-685 - - -	4	6.8	0.5	5.0	6.0	6	9	9	5.5	B
CWR11C-106 - - -	4	10.0	0.5	5.0	6.0	6	9	9	4.0	B
CWR11C-156 - - -	4	15.0	0.6	6.0	7.2	6	9	9	3.5	B
CWR11C-336 - - -	4	33.0	1.3	13.0	15.6	6	9	9	2.2	C
CWR11C-686 - - -	4	68.0	2.7	27.0	32.4	6	9	9	1.1	D
CWR11C-107 - - -	4	100.0	4.0	40.0	48.0	8	12	12	0.9	D
CWR11D-155 - - -	6	1.5	0.5	5.0	6	6	9	9	8.0	A
CWR11D-225 - - -	6	2.2	0.5	5.0	6	6	6	9	8.0	A
CWR11D-335 - - -	6	3.3	0.5	5.0	6	6	9	9	8.0	A
CWR11D-475 - - -	6	4.7	0.5	5.0	6	6	9	9	5.5	B
CWR11D-685 - - -	6	6.8	0.5	5.0	6	6	6	9	4.5	B
CWR11D-106 - - -	6	10.0	0.6	6.0	7.2	6	9	9	3.5	B
CWR11D-156 - - -	6	15.0	0.9	9.0	10.8	6	6	9	3.0	C
CWR11D-226 - - -	6	22.0	1.4	14.0	16.8	6	9	9	2.2	C
CWR11D-476 - - -	6	47.0	2.8	28.0	33.6	6	6	9	1.1	D
CWR11D-686 - - -	6	68.0	4.3	43.0	51.6	6	9	9	0.9	D
CWR11F-105 - - -	10	1.0	0.5	5.0	6.0	4	6	6	10.0	A
CWR11F-155 - - -	10	1.5	0.5	5.0	6.0	6	6	9	8.0	A
CWR11F-225 - - -	10	2.2	0.5	5.0	6.0	6	9	9	8.0	A
CWR11F-335 - - -	10	3.3	0.5	5.0	6.0	6	9	9	5.5	B
CWR11F-475 - - -	10	4.7	0.5	5.0	6.0	6	9	9	4.5	B
CWR11F-685 - - -	10	6.8	0.7	7.0	8.4	6	9	9	3.5	B
CWR11F-156 - - -	10	15.0	1.5	15.0	18.0	6	6	9	2.5	C
CWR11F-336 - - -	10	33.0	3.3	33.0	39.6	6	9	9	1.1	D
CWR11F-476 - - -	10	47.0	4.7	47.0	56.4	6	9	9	0.9	D
CWR11H-684 - - -	15	0.68	0.5	5.0	6.0	4	6	6	12.0	A
CWR11H-105 - - -	15	1.0	0.5	5.0	6.0	4	6	6	10.0	A
CWR11H-155 - - -	15	1.5	0.5	5.0	6.0	6	9	9	8.0	A
CWR11H-225 - - -	15	2.2	0.5	5.0	6.0	6	9	9	5.5	B
CWR11H-335 - - -	15	3.3	0.5	5.0	6.0	6	8	9	5.0	B
CWR11H-475 - - -	15	4.7	0.7	7.0	8.4	6	9	9	4.0	B
CWR11H-106 - - -	15	10.0	1.6	16.0	19.2	6	8	9	2.5	C
CWR11H-226 - - -	15	22.0	3.3	33.0	39.6	6	8	9	1.1	D
CWR11H-336 - - -	15	33.0	5.3	53.0	63.6	6	9	9	0.9	D
CWR11J-474 - - -	20	0.47	0.5	5.0	6.0	4	6	6	14.0	A
CWR11J-684 - - -	20	0.68	0.5	5.0	6.0	4	6	6	12.0	A
CWR11J-105 - - -	20	1.0	0.5	5.0	6.0	4	6	6	10.0	A
CWR11J-155 - - -	20	1.5	0.5	5.0	6.0	6	9	9	6.0	B
CWR11J-225 - - -	20	2.2	0.5	5.0	6.0	6	8	9	5.0	B
CWR11J-335 - - -	20	3.3	0.7	7.0	8.4	6	9	9	4.0	B
CWR11J-475 - - -	20	4.7	1.0	10.0	12.0	6	8	9	3.0	C
CWR11J-685 - - -	20	6.8	1.4	14.0	16.8	6	9	9	2.4	C
CWR11J-156 - - -	20	15.0	3.0	30.0	36.0	6	8	9	1.1	D
CWR11J-226 - - -	20	22.0	4.4	44.0	52.8	6	9	9	0.9	D

See footnote at end of table.

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TABLE I. Style CWR11 characteristics - Continued.

PIN 1/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR11K-334 - - -	25	0.33	0.5	5.0	6.0	4	6	6	15.0	A
CWR11K-474 - - -	25	0.47	0.5	5.0	6.0	4	6	6	14.0	A
CWR11K-684 - - -	25	0.68	0.5	5.0	6.0	4	6	6	7.5	B
CWR11K-105 - - -	25	1.0	0.5	5.0	6.0	4	6	6	6.5	B
CWR11K-155 - - -	25	1.5	0.5	5.0	6.0	6	8	9	6.5	B
CWR11K-225 - - -	25	2.2	0.6	6.0	7.2	6	9	9	3.5	C
CWR11K-335 - - -	25	3.3	0.9	9.0	10.8	6	8	9	3.5	C
CWR11K-475 - - -	25	4.7	1.2	12.0	14.4	6	9	9	2.5	C
CWR11K-685 - - -	25	6.8	1.7	17.0	20.4	6	9	9	1.4	D
CWR11K-106 - - -	25	10.0	2.5	25.0	30.0	6	8	9	1.2	D
CWR11K-156 - - -	25	15.0	3.8	38.0	45.6	6	9	9	1.0	D
CWR11M-104 - - -	35	0.10	0.5	5.0	6.0	4	6	6	24.0	A
CWR11M-154 - - -	35	0.15	0.5	5.0	6.0	4	6	6	21.0	A
CWR11M-224 - - -	35	.22	0.5	5.0	6.0	4	6	6	18.0	A
CWR11M-334 - - -	35	.33	0.5	5.0	6.0	4	6	6	15.0	A
CWR11M-474 - - -	35	.47	0.5	5.0	6.0	4	6	6	10.0	B
CWR11M-684 - - -	35	.68	0.5	5.0	6.0	4	6	6	8.0	B
CWR11M-105 - - -	35	1.0	0.5	5.0	6.0	4	6	6	6.5	B
CWR11M-155 - - -	35	1.5	0.5	5.0	6.0	6	8	9	4.5	C
CWR11M-225 - - -	35	2.2	0.8	8.0	9.6	6	8	9	3.5	C
CWR11M-335 - - -	35	3.3	1.2	12.0	14.4	6	8	9	2.5	C
CWR11M-475 - - -	35	4.7	1.7	17.0	20.4	6	8	9	1.5	D
CWR11M-685 - - -	35	6.8	2.4	24.0	28.8	6	9	9	1.3	D
CWR11N-104 - - -	50	.10	0.5	5.0	12	6	8	8	22.0	A
CWR11N-154 - - -	50	.15	0.5	5.0	6.0	4	6	6	17.0	B
CWR11N-224 - - -	50	.22	0.5	5.0	6.0	4	6	6	14.0	B
CWR11N-334 - - -	50	.33	0.5	5.0	6.0	4	6	6	12.0	B
CWR11N-474 - - -	50	.47	0.5	5.0	6.0	4	6	6	8.0	C
CWR11N-684 - - -	50	.68	0.5	5.0	6.0	4	6	6	7.0	C
CWR11N-105 - - -	50	1.0	0.5	5.0	6.0	4	6	6	6.0	C
CWR11N-155 - - -	50	1.5	0.8	8.0	9.6	6	8	9	4.0	D
CWR11N-225 - - -	50	2.2	1.1	11.0	13.2	6	8	9	2.5	D
CWR11N-335 - - -	50	3.3	1.7	17.0	20.4	6	9	9	2.0	D
CWR11N-475 - - -	50	4.7	2.4	24.0	28.8	6	9	9	1.5	D

1/ Complete PIN shall include additional symbols to indicate termination finish, capacitance tolerance, product level, and, if applicable, surge current option letter. If optional surge current is not required, the last "-" shall be deleted.

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Army - CR
Navy - EC
Air Force - 85
DLA - CC

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Navy - AS, MC, OS, SH
Air Force - 19
Other – NA

Preparing activity:
Army - CR

Agent:
DLA - CC

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